

Gianmarco Santoro

AI Engineer focusing on **Computer Vision**, **Robotics**, and applied **ML**

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◆ Experience

Applied AI Advisor | *Startups (Confidential)* | 2025 – Present

- Supporting early-stage startups helping industrial SMEs **adopting AI to automate workflows** and improve processes. Case study:
 - Rapid prototyping of lightweight **proof-of-concept** system to **automate CNC programming** workflows. Proposed an architecture to extract features from CADs, G-codes and technical sheets to **fine-tune a compact LLM** (e.g.T5-small) for task-specific code generation. Analysed multiple solutions including GNN/transformer models, and an alternative computer-vision approach (VLMs).

DeepFake Detection | **Computer Vision - Machine Learning Engineer** | *MICC* | *Florence, Italy* | 2023 - 2024

- Implemented **Out-of-Distribution (OOD) Detection** on *FaceForensics++* dataset to identify AI-generated images. A 3-step pipeline:
 - Extracted Self-Attention heatmaps via **Vision Transformer**, pre-trained on **face recognition**.
 - Exploited heatmaps: trained a **Convolutional Autoencoder** on real ones.
 - OODD: tested real vs fake reconstruction error, achieved **60% AUROC across 5 forgeries**, showing transfer learning capabilities.

Smart Center | **SW & System Integration Engineer** | *Daikin Applied Europe SpA* | *Rome, Italy* | 2021 - 2022

- Worked on cloud-based HVAC monitoring, system integration, and services:
 - Automated PLC** mapping and alarm selection via **Python**, reducing manual configuration time by ~95%.
 - Built real-time **IoT sensor-to-cloud temperature control system** integrating 21 Modbus sensors, removing 100% manual setup.
 - Aggregated data via web API to generate 2000+ reports/month, improving issues detection and reducing workload by ~80%.

Wireless Charging System | **Operations Research Scientist** | *Enermove Srl* | *Turin, Italy* | 2020 - 2021

- Optimized electric vehicles wireless charging system topology via **Genetic Algorithms**, improving energy efficiency-to-cost ratio.

Chip Extrusion | **Manufacturing Researcher** | *Institut für Umformtechnik und Leichtbau* | *Dortmund, Germany* | 2018

- Experimented with the **aluminum recycling process** by direct extrusion of mixed waste alloys (DoE). Compared measured properties with FEM and numerical results. Assessed **industrial feasibility** identifying process bottlenecks: logistic issues and product defects.

◆ Education

Data Science & Statistical Learning | *Post-M.Sc. at UniFi & IMT Lucca – JoTTO excellence Uni* | 2023 - 2024 | 110/110 cum laude

- Some topics: Machine Learning, DL, RL, Statistics, **Optimization**, GIS, NLP, Graph Theory. • Projects, [Website](#) for more:
 - Developed **Deep Learning images classifier** via **CNNs** models (PyTorch), achieving an accuracy of 70.4% on the test set.
 - Built **ML housing price pipeline**: K-Fold Cross-Validation, **Grid Search** on polynomial regression degree improving MAE by 12%.
 - Predicted salary** on selected features using **CART** and other models. Reached 83.8% accuracy with Gradient Boosting.
 - Analyzed Defibrillators **geographical distribution** (GIS): found **clustering** aligned with population density and seasonal peaks.

Mechanical Engineering | *M.Sc. at PoliTo – Top 30 in Mechanical Engineering, QS Rankings* & *B.Sc. at UniBo* | 104/110

- Specialized in **Automation**. Some extra projects: • Automation and **mechatronics** labs: PLC, PID controls, pneumatics, mechanisms.
- Designed a reduction **gearbox**. • Developed a home **heating system**. • Built a **numerical model** for metal flue heat exchange.

◆ **Side Projects** | 2022 - Present

- Conducted Market Research, Javelin Board and SWOT analysis, Lean Startup Method on PoC/MVPs: **Project-Driven** Work Community; **Local Food** Logistics Platform; **Food-based** Social Media; **Selected Products** Referral Platform.

◆ **Skills:** **Languages:** • **Italian:** C2 - Native • **English:** C1 - Professional • **German:** A2 - Basic

AI & SW: Python, PyTorch, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, GIT, R & Rstudio, Matlab & Simulink

Web & Tools: HTML, CSS, VSCode, Conda, Jupyter/Colab, GitHub/GitLab, MS-Office (Excel), Google-Suite (forms), Notion, Canva

Industrial: AutoCAD, SolidWorks, CATIA, HyperMesh/Extrude, MSC-Adams/Apex, TCP/IP, mRemoteNG, Modbus, qModMaster

Basic exposure & Extra: Bash, zsh, CUDA, C, JS, PLC Siemens-Climatix, SSH, Matlab Optim. Toolbox, LaTeX